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Retraction: Angiotensin-converting enzyme I/D gene polymorphism affects early cardiac response to professional training in young footballers

The following article has been retracted at the request of the Journal Editor and Publisher:

Saber-Ayad MM, Nassar YS and Latif IA. Angiotensin-converting enzyme I/D gene polymorphism affects early cardiac response to professional training in young footballers. *Journal of the Renin-Angiotensin-Aldosterone System* 2014; 15(3): 236–242. DOI: 10.1177/1470320312471150.

Sage was alerted to a PubPeer post highlighting potential duplication across figures in the article.

Concerns were raised about the following figures in the article:

1. In Figure 1, lane 1: 'DNA marker' appears to be duplicated in Figure 2, lane 3: 'DNA marker.'
2. In Figure 1, lanes 3: 'Homozygous II genotype', 4 and 5: 'I genotype' appear identical.
3. In Figure 1, lanes 2: 'Homozygous D/D genotype', 4 and 'D genotype' appear identical.
4. In Figure 2, sample lanes 1 and 2 appear highly similar.

The authors confirmed that the 'DNA marker' appeared in both Figure 1 and 2. The authors were unable to provide the uncropped raw images underlying these figure panels and were unable to resolve the remaining concerns raised.

Due to the lack of raw images and unresolved concerns about image integrity that call into question the validity of the findings the Journal Editor and Sage retracts the article.

The authors did not respond to the decision to retract.

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RETRACTED: Angiotensin-converting enzyme I/D gene polymorphism affects early cardiac response to professional training in young footballers

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Abstract

Background: The renin–angiotensin–aldosterone system plays a role in physiological and pathological responses of the heart to both static and dynamic exercise. Previous studies showed that the level of angiotensin II is determined by the angiotensin-converting enzyme insertion/deletion (ACE I/D) polymorphism.

Aim: We aimed in this study to determine the effect of ACE I/D gene polymorphism on the extent of functional and structural cardiac changes in response to one year of professional football training in young footballers.

Methods and results: We studied 68 young male football players and a comparable control group. Besides medical history and clinical examination, 12 lead ECG and transthoracic 2D echocardiography examination were performed. Genotyping of ACE was analyzed using PCR-based technique. There was no statistically significant difference in distribution of genotypes among athletes compared with control subjects. D allele showed a graded effect on both EF (73.55, 67.5 and 60.2%, p=0.03) and PASP (37.6, 26.1 and 21.39 mmHg, p=0.02) in DD, ID and II subjects, respectively.

Conclusion: Early cardiac changes in young footballers can be affected by ACE I/D polymorphism. There is a summative effect of the D allele in increasing EF and PASP in response to professional football training.

Keywords

ACE I/D, gene polymorphism, exercise physiology, left ventricular mass, echocardiography

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Introduction

Strong evidence exists for racial differences in cardiac response to certain physiological and pathological stimuli. For instance, Afro-Caribbean patients with systemic hypertension show a significantly higher mean left ventricular diameter compared with hypertensive Caucasians.[1] Very little is known about African athletes, even less about Egyptian.

Physiological adaptation to regular static and dynamic training has been previously reported. Intensive and prolonged physical training leads to cardiovascular adjustments that allow for an exceptional physical performance of athletic heart.[2–6]

Echocardiographic studies documented typical morphologic changes in the left ventricle (LV), such as the increase in end diastolic diameter (EDD), myocardial wall thickness and myocardial mass, which correspond to eccentric hypertrophy.[7] In contrast, only limited data are available on the right ventricle (RV), because of its complex shape and marked trabeculations, which may exceed the validity and reliability of echocardiography.[8]

Plasma and tissue concentration of angiotensin-converting enzyme (ACE) are determined by the ACE gene (chromosome 17q23). An important polymorphism in this gene comprises the insertion (I) or deletion (D) of a 287-bp in intron 16 of the gene. Studies showed that the homozygous DD genotype is associated with increased serum ACE level[9] and cardiac ACE activity.[10]

In view of its potential effect, we aimed in this study to determine the effect of ACE I/D gene polymorphism on the extent of functional and structural cardiac changes in

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response to one year of professional football training in young Egyptian athletes.

Methods

Subjects

Football players trained for selection in the Egyptian Young World Cup National team were invited to join this study. During the follow-up year 68 out of 75 young athletes (participation 90.7%) aged 17–21 years with football as their main training modality gave their written informed consent to take part in the study (according to the instructions and regulations of the Medical Ethics Committee in Cairo University). The control group consisted of 100 age-, sex-, height- and weight-matched, non-smoking, sedentary, third year students studying at the medical school, Cairo University. None of them exercised for more than 30 minutes per week. After being given complete information about the study, participants signed consent. All athletes and control subjects were asymptomatic Egyptians coming for different parts of the country. They are free from cardiac disease. None of them was known to be using anabolic steroids or any other medications that may cause cardiac hypertrophy.

Study protocol

The study protocol was approved by the ethics Committee of Kasr El-Eini Medical School, Cairo University. The subjects participated in 2D-echocardiographic and Doppler study at rest.

Echocardiographic measurements

Echocardiographic and Doppler studies were performed using an Esaote MyLab30 Echomachine. Echocardiographic examination was performed with the subject at rest, lying in left lateral decubitus position. Both standard parasternal views (long and short axis) and apical views (four-, two- and three-chamber) were obtained. An integrated M-mode and two-dimensional study was done, consistent with the recommendations of the American Society of Echocardiography[11] and the European Association of Echocardiography recommendations.[12] The LV mass was calculated using the formula of Devereux.[13] LV wall tension (WT) was calculated using the following formula:[14] WT (dyn/cm) = 1.33 × systolic blood pressure × (LV end systolic diameter)/2. Measurements were quoted with body surface area (BSA) calculated with the formula of Dubois and Dubois.[15] Each measurement was made on three consecutive cardiac cycles and an average of the three values was considered. The static and dynamic images were reviewed by another echocardiographer for the purpose of excluding criteria for pathological hypertrophy in the athlete group.

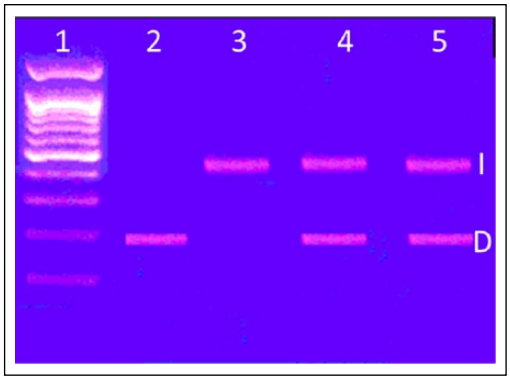

Figure 1. Angiotensin-converting enzyme insertion/deletion genotyping (first PCR reaction). Lane 1: DNA marker. Lane 2: Homozygous D/D genotype. Lane 3: Homozygous II genotype. Lanes 4, 5: I/D genotype.
D: deletion; I: insertion.

Genotyping of the ACE gene I/D polymorphism

After obtaining informed consent for inclusion into the study from the parents, genomic DNA was extracted from 200 µl of whole blood with EZ-10 Spin column DNA extraction minipreps kit (Biobasic Company, Canada). The I/D polymorphism of the ACE gene was determined according to the method of Rigat et al.[16] The sequences of the sense and antisense primers were 5′-CTG GAG ACC ACT CCC ATC CTT TCT-3′ and 5′-GAT GTG GCC ATC ACA TTC GTC AGA T-3′, respectively (prepared by AM-Egypt). PCR was performed in a final volume of 50 µl that contained 25 µl master mix, ≈500 ng of genomic DNA, 12.5 pmol of each primer and 5% dimethylsulfoxide (DMSO). Amplification was performed using a Gene Amp PCR system 2400–Perkin Elmer. Samples were denatured for 1 min at 94°C and then cycled 30 times through the following steps: 45 s at 94°C, 1 min at 62°C, 1 min at 72°C. PCR products were electrophoresed in 1.6% agarose gel and visualized directly with ethidium bromide staining. The I allele was detected as a 490-bp band, and the D allele was detected as a 190-bp band (Figure 1). DMSO was included in the PCR to prevent underestimation of heterozygotes and overestimation of the DD genotype. A second PCR amplification was performed for each DD type with a primer pair that recognizes an insertion-specific sequence (5′- TGG GAC CAC AGC GCC CGC CAC TAC-3′; 5′-TCG CCA GCC CTC CCA TGC CCA TAA-3′), with identical PCR conditions except for an annealing temperature of 67°C and the absence of 5% DMSO[17] (Figure 2).

Statistical analysis

The allele distribution of the ACE gene I/D polymorphism was tested for Hardy–Weinberg equilibrium in footballer

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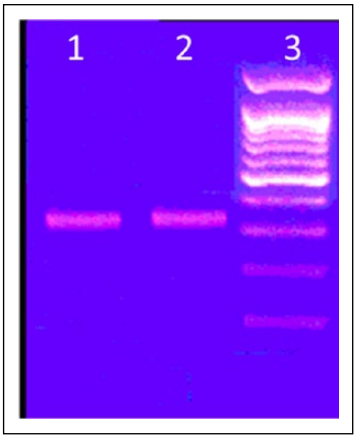

Figure 2. Angiotensin-converting enzyme insertion/deletion genotyping (second PCR reaction). All samples that showed D/D genotypes in the first reaction will undergo a second reaction using different primers. The reaction recognizes insertion-specific sequence. Lanes 1 and 2: insertion allele was recognized in samples that showed D/D genotype in the first PCR reaction, that is, samples in Lane 1 and 2 turned out to be I/D genotype (no bands will result if the sample is D/D genotype). Lane 3: DNA marker.

and control groups. A computer software package (SPSS), version 15, was used in the analysis. For quantitative variables, mean and standard deviation are presented. Frequency and percentage are presented for qualitative variables. Chi square/Fisher's Exact were used to estimate differences in qualitative variables. Analysis of variance (ANOVA) was used to measure differences among the three genotypes. An alpha error level of $p<0.05$ was considered as statistically significant.

Results

Football players recruited for the National Youth Football Team were selected from amateur footballers who were used to participating in irregular football playing for 3–4.5 h/week for 5±1 years prior to joining the National Team. After their recruitment, they were under a regular training program including both endurance and strength components for 13 h/week (2 h static exercises + 11 h dynamic training/week).

The distribution of ACE gene I/D polymorphism in both footballers and control groups is shown in Table 1. Different genotypes as well as the distribution of either I or D alleles were analyzed. There are no statistically significant differences between the two groups.

Table 2 shows the clinical characteristics of the study group. Both footballers and control subjects are matched regarding age, height, weight, body surface area and body mass index.

Table 3 shows the electrocardiographic measurements of football players and the control group. Differences between the two groups were not statistically significant apart from the heart rate.

Echocardiographic examination showed statistically significant differences between both groups. The footballer group showed significantly higher LV mass ($p = 0.027$) and ejection fraction (EF) ($p = 0.02$). Footballers showed lower WT and lower pulmonary artery systolic pressure (PASP), $p<0.001$ (Table 4).

We focused on those echocardiographic parameters that showed significant differences between the two groups and studied whether there is an effect of ACE I/D polymorphisms (Table 5).

EF showed significant differences among DD, ID and II genotypes, with DD genotype having the highest EF (73.55%), with ID 67.5% and II 60.2%. There was also significantly higher PASP in footballers with DD genotype (37.6 mmHg) as compared with II genotype (21.39 mmHg). This was reflected on the right ventricular diameter (RVd), (25.95±3.5 and 22.38±2.9 mm in DD and II genotypes, respectively). Other parameters showed no statistically significant differences although the effect of DD allele appeared to be summative. Such an effect of D allele was not observed among different genotypes in the control group. Although EF was highest in the DD genotype (63.72%), this was not statistically significant from the other two groups (60.1% and 59.8% in ID and II sub-groups, respectively, $p = 0.18$). Similarly, mean RVd was 22.8, 22.39 and 22.52 mm in DD, ID and II sub-groups, respectively ($p = 0.38$) and mean PASP was 15.21, 15.1 and 15.8 mmHg ($p = 0.67$).

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Table 1. Distribution of the ACE gene I/D polymorphism in a sample of footballers compared with controls.

	Footballer players (n = 68)	Control subjects (n = 100)	Fisher p
Genotype			
DD	26 (38.2%)	39 (39%)	NS
ID	34 (50%)	49 (49%)	NS
II	8 (11.8%)	12 (12%)	NS
Alleles			
D	60 (58.8%)	88 (59.1%)	NS
I	42 (41.2%)	61 (40.9%)	NS

I: insertion; D: deletion; NS: non-significant.

Table 2. Clinical characteristics of the one-year professionally trained football players and sedentary control subjects.

Variable	Football players (*n*=68)		Control subjects (*n*=100)		*p* value
	Mean±SD	Range	Mean±SD	Range	
Age (years)	18.82±1.6	17–21	19.64±2.3	17–22	0.12
Height (cm)	176.15±5.5	168–190	177.0±6.5	162–187	0.56
Weight (kg)	72.29±5.6	62–87	74.82±13.9	57–90	0.37
BSA (m^2)	1.87±0.1	1.72–2.1	1.9±0.2	1.7–2.3	0.31
BMI (kg/m^2)	22.9±1.2	23.1–23.6	23.7±3.6	20.5–22.3	0.27
Professional training	13±1 (h/week) for one year*	12–14	none		
Heart rate (bpm)	68.88±5.0	41–90	77.43±6.0	52–98	<0.001
Systolic BP (mmHg)	117.79±6.5	108–126	126.43±17	110–150	0.008
Diastolic BP (mmHg)	74.41±5.6	64–94	76.79±12.5	70–100	0.32

Values are mean ± standard deviation (SD) and range for 34 football players and 28 control subjects. BSA: body surface area; BMI: body mass index; bpm: beats per minute; BP: blood pressure.
*Weekly training includes 11 h of dynamic and 2 h of static training.

Table 3. Electrocardiographic measurements of one-year trained football players and sedentary control subjects.

Variable	Football players (*n*=68)	Control subjects (*n*=100)	*p* value
Heart rate (bpm)	68.88±5.0	77.43±6.0	<0.001
PR interval (s)	0.15±0.02	0.14±0.03	0.55
QRS complex width (s)	0.1±0.06	0.08±0.06	0.21
QTc (ms)	432±28	424±22	0.14

Values are mean ± standard deviation.
bpm: beats per minute; QRS: ECG wave representing ventricular depolarization; QTc: QT interval corrected to the heart rate.

Table 4. Echocardiographic measurements in professionally trained football players and sedentary control subjects.

Variable	Football players (*n* = 68)	Control subjects (*n* =100)	*p* value
Aortic root (mm)	27.9±3.2	26.3±4.1	0.1
LVEDD (mm)	47.9±5.9	46.9±6.5	0.4
LVESD (mm)	31.9±7.2	34.2±5.0	0.12
PWT (mm)	10.96±2.8	11.4±2.3	0.19
SWT (mm)	12.5±3	11±2.4	0.09
Sum of LV wall thickness (mm)	23.55±3.7	23.02±3.0	0.55
LV mass (g)	243.61±34.2	202.54±32.6	0.027
Indexed LV mass (g/m^2)	129.88±20.3	107±18.1	0.09
Wall tension (dyn/cm)	262.13±15.8	294.52±18.4	0.022
EF (%) (by Simpson's Rule)	68.95 ±5.4	61.45±6.7	0.02
RVDd (mm)	24.77±3.84	22.57±4.85	0.1
PASP (mmHg)	29.73±7.2	15.23±3.4	<0.001

Values are mean ± standard deviation.
LVEDD: left ventricular end diastolic diameter; LVESD: left ventricular end systolic diameter; PWT: posterior wall thickness; SWT: septal wall thickness; EF: ejection fraction; RVDd: right ventricular diameter in diastole; PASP: pulmonary artery systolic pressure.

Table 5. Distribution of ACE I/D genotypes in relation to echocardiographic parameters that showed significant difference between football and control groups.

Echocardiographic parameter	Genotype			*p* value (ANOVA)
	DD (*n*=26)	ID (*n*=34)	II (*n*=8)	
LV mass (g)	254.33±38.60	240.94±32.8	220.25±34.1	0.69
Indexed LV mass (g/m^2)	131.80±19.74	130.03±16.3	123.04±24.23	0.82
Wall tension (dyn/cm)	281.69±18.45	261.74±12.35	257.02±12.45	0.46
EF (%)	73.55±7.5	67.5±4.1	60.2±5.4	0.03
RVDd (mm)	25.95±3.5	24.43±3.0	22.38±2.9	0.04
PASP (mmHg)	37.6 ±2.3	26.1±2.5	21.39±4.2	0.02

LV: left ventricle; EF: ejection fraction; RVDd: right ventricular diameter in diastole; PASP: pulmonary artery systolic pressure.

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Discussion

The role of the renin–angiotensin–aldosterone system in promoting physiological and pathological responses in the heart has been established. Previous investigations showed varied and sometimes paradoxical results regarding the effect of I/D gene polymorphism in response to physical training.[18,19]

The ACE DD genotype is associated with increased circulating ACE levels, which are generally two times as high as those found for the II genotype; ID heterozygotes are associated with intermediate ACE levels.[9]

Regarding genotype frequency, both footballers and controls showed similar distribution with ID genotype being the most common (50% in footballers and 49% in controls), followed by DD genotype (38.2% and 39%) and II, which showed the least frequency (11.8% and 12%). Our results are comparable to previous studies made in Egyptian populations.[20] Settin et al. showed different results (DD=47.5%, ID 47.5% and II 5%).[21] This discrepancy may be due to the heterogeneity of the Egyptian population and the relatively small number of subjects in both investigations.

In contrast to our investigation, a number of studies showed an association between ACE D allele and endurance athletes in different populations.[22,23]

The distinction between physiological and pathological conditions in athlete's heart is of great importance, as the latter may result in critical implications for the athlete's lifestyle and professional career.

In our study, we paid particular attention to the previously described criteria differentiating athlete's heart from hypertrophic cardiomyopathy (HCM).[24,25] Family history was explored. All echocardiographic images included in the study were independently interpreted by two echocardiographers to obtain the measurements; criteria of HCM were cautiously sought for in all cases, in particular posterior and septal wall thickness. Elite junior athletes have greater absolute LV wall thickness, rarely exceeding 12mm, and always with chamber enlargement.[26]

LV mass is significantly higher in athletes compared with that of control subjects. As a combined endurance and strength training, football training usually exposes the athlete to a high pressure load that leads to a marked increase in LV wall thickness and dimensions (both end systolic and end diastolic).[27] Further follow-up of our study population will help highlight stepwise adaptive changes that sequentially occur with continuing professional training.

Supportive of our results, previous studies showed significant increase in LV wall thickness, diameter and mass in athletes compared with controls in both Caucasians[28] and Africans.[29]

The effect of ACE I/D gene polymorphism on LV mass in our study is obvious; there is a summative effect of the presence of D allele, with DD genotype carriers showing the highest LV mass, II showing the least and ID showing an intermediate value. Previous studies showed similar results in response to 10-week physical training in male Caucasians[19,30] as well as in strength-trained elite athletes.[31]

It should be noted that such effect is a dynamic rather than a static one. In concordance with our results, some studies showed the effect of this gene polymorphism on the LV mass in response to exercise,[19] whereas others negated the association of ACE I/D gene polymorphism and LV mass in more than 2000 subjects from the Framingham Heart Study of non-athletes.[18]

WT reflects the effect of the preload on the myocardium. In our study the WT was significantly lower in athletes as compared with controls, whereas a significant increase in EF in the former was noticed. Calculation of both parameters depends on the end systolic diameter. Our results are supported by previous studies investigated Caucasian athletes.[19]

Our study showed significant difference in both WT and EF among the three genotypes, with cumulative effect of D allele. Increased angiotensin II (Ang II) may lead to stiffness of the cardiac muscle with resultant increase in WT. Supporting our results, previous studies showed that ACE DD and AT1R 1166 CC genotypes, concomitantly present, increase the predisposition to diastolic heart failure.[32]

Few studies investigated the early adaptive right-sided changes occurring in athlete's heart, especially in the young. Our results showed a significant increase in PASP in athletes in comparison with control subjects. In addition, a significant increase in RV diameter followed.

High levels of physical activity in adults are associated with greater RV mass and volume independent of the associations with LV mass due to exercise-associated RV remodeling.[33]

McQuillan et al. observed a wide spectrum of PASP in normal subjects. They concluded that PASP varies independently with age, BMI, male sex, LV posterior wall thickness and LV ejection fraction.[34]

A recent study has shown that endurance-trained athletes have a significantly higher PASP in comparison with strength-trained athletes and control subjects.[35] The study concluded that the upper physiologic limit of PASP in endurance athletes may reach up to 40mmHg. Although this study was evaluating highly trained athletes, it can give an explanation to our unexpected results of elevated PASP.

In our study, footballers with DD genotype showed an increase in both LV mass and PASP. These findings can be considered a physiologic consequence of long-term and intensive exercise conditioning. The reduced peripheral resistance that accompanies isotonic aerobic sports can determine venous overload and consequent LV preload, thus inducing greater LVEDD and LV stroke volume.[36] Therefore, the higher PASP we noticed in this study may be explained by the proportional increase in LV stroke volume.

In our study, ACE gene polymorphism showed an effect on both sides of the heart. The effect was significant on the structure and function of the right side. DD genotype showed a significantly higher PASP and RV diameter compared with ID and II genotypes.

In concordance with our results, Kanazawa et al. showed an association of ACE gene polymorphism with an increase in mean pulmonary artery pressure as well as pulmonary vascular resistance in response to exercise in chronic obstructive pulmonary disease.[37] Similar results were reported by Tkácová et al.[38]

The lungs are metabolically active organs that produce Ang II and endothelin among other peptides.[39] Pulmonary vascular tone is under the control of both vaso-active peptides.

In addition to its vaso-active property, Ang II also induces proliferation and inflammation of the pulmonary vasculature that contributes to the rise of pulmonary artery pressure in both primary and secondary pulmonary hypertension (e.g. in patients with chronic obstructive airway disease).[40,41] Recently, ACE2 enzyme was shown to be a negative regulator of the angiotensin system (counter-regulatory homolog of ACE). It metabolizes Ang II to a putative protective peptide angiotensin (1-7) and leads to vasodilatation and anti-proliferative effect.[39] Therefore, it is expected to be of therapeutic benefit in treating pulmonary hypertension.[40]

The results of our study support the direct relationship between the higher levels of ACE (and subsequently Ang II) and the PASP, which showed a gradual rise according to the number of D alleles.

In a recent study by Ventetuolo et al., the researchers showed that RAS manipulation by using ACE inhibitors or angiotensin receptor blockers was associated with lower RV mass and smaller RV volumes among African Americans, even after adjustment for respective LV measures and spirometry. This gives indirect evidence that RAS manipulation may uniquely affect RV morphology independent of its effect on the LV.[41] Further studies should investigate the assumption that Ang II affects the pulmonary vasculature earlier and more significantly than it does the systemic.

The balance between Ang II and Ang 1-7 seems to be disturbed in individuals with DD genotype in favor of Ang II. The D allele may be in linkage disequilibrium with another causative variant in ACE2 gene that decreases the enzyme level or activity, and suppresses the production of Ang 1-7 in the lungs. Further studies are required to elucidate the role of this gene as a potential factor in determining vascular resistance.

In addition, the preferential effect of Ang II/ACE2 on the right side of the heart may be related to the higher enzymatic activity of the pulmonary circulation as compared with the systemic.

To the best of our knowledge, this is the first study that investigated the role of ACE I/D gene polymorphism in an African population, concerning early cardiac response to professional training. Follow-up of our study population may throw light on the significance and explanation of the study results.

Limitations

We conducted the study on a relatively small number of athletes and controls. We did not use relatively recent techniques (tissue Doppler, 3D echocardiography and magnetic resonance). We did not measure the volume or the function of the right ventricle. Investigating the genes of the renin–angiotensin–aldosterone system included only ACE I/D polymorphism.

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Conflict of interest

None declared.

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